

MINOR-2 PROJECT
SYNOPSIS on
Deepfake Detection System

Submitted By:

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Under the guidance of
Dr. Bhavana Kaushik
(Assistant professor at School of Computer Science)



School of Computer Science

University of Petroleum & Energy Studies, Dehradun

Dehradun-248007

Jan-Apr 2025

Approved By

Project Guide

Cluster Head

(1) pytorch.
(2) Model explanation
(3) Data set not fixed.

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Introduction

Deepfakes, especially face-swapped ones, are very dangerous as they manipulate facial features in videos for malicious purposes like misinformation, fraud, and defamation. Artificial Intelligence (AI) and Machine Learning (ML) have been rapidly advancing, making deepfakes more sophisticated, thus it is challenging to differentiate between real and fake videos. This project will develop an AI/ML-based solution that can detect face-swapped deepfake videos by analysing video frames and identifying inconsistencies. The system will be able to differentiate between real and altered videos with high accuracy using deep learning techniques like Convolutional Neural Networks (CNNs).

Deepfake detection should ensure that what appears in your digital media actually happens and safeguards people from ID fraud. Thus, it controls the spread of misinformation across channels like social networks and news corporations.

Problem Statement

Deepfake technology has evolved significantly, making it challenging to identify manipulated videos using traditional detection methods. Face-swapped deepfakes, generated using advanced AI models like Generative Adversarial Networks (GANs), have become highly realistic and difficult to detect.

Challenges in detecting face-swap deepfakes:

- Traditional detection methods are ineffective against new, adaptive deepfake techniques.
- Deepfake videos can bypass simple forensic detection methods.
- Continuous advancements in deepfake generation techniques make detection a moving target.
- A robust AI-based detection solution is required to identify inconsistencies in facial features, expressions, and lighting conditions across frames in a video.

Our project focuses on building a reliable, scalable, and efficient AI/ML-based deepfake detection system to tackle these challenges.

Literature Review

Reference	Methods	Year of Research
[1]	A large-scale benchmark dataset for deepfake detection, widely used for training deepfake detection models.	2019
[2]	A dataset designed to test the effectiveness of various deepfake detection models.	2020
[3]	An open-source implementation of deepfake detection using deep learning techniques.	2023
[4]	A real-time deepfake detection system that analyzes subtle biological signals in human faces to detect deepfakes.	2022
[5]	Studies have highlighted the limitations of existing models and the need for continual improvements in detection algorithms.	2024

Objectives

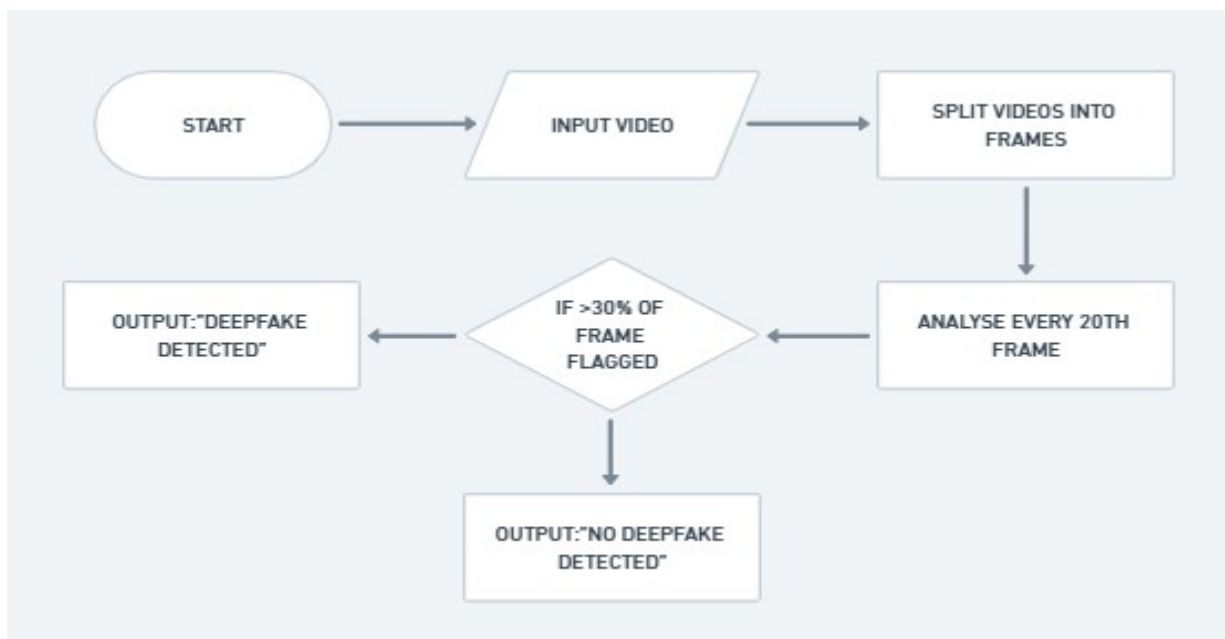
The primary objectives of this project are:

- To develop an AI/ML-based model capable of detecting face-swapped deepfake videos with high accuracy.
- To analyze facial inconsistencies, expression mismatches, and pixel-level anomalies in video frames.
- To implement a frame-by-frame analysis technique for identifying deepfake characteristics.
- To explore different deepfake detection models and enhance accuracy using hybrid approaches.
- To build a user-friendly interface that provides clear detection results with detailed explanations.
- To ensure scalability by experimenting with multiple deepfake detection techniques and integrating open-source solutions.

Methodology

Technical Approach:

1. **Preprocessing:** The input video is divided into frames, and key frames are extracted for analysis.
2. **Feature Extraction:** CNN-based deepfake detection models analyze facial expressions, texture inconsistencies, and lighting differences.
3. **Frame Analysis:** Every 20th frame is analyzed for deepfake characteristics.
4. **Detection Criteria:** If more than 30% of analyzed frames show anomalies, the video is flagged as a deepfake.
5. **Prototyping:**
 - o **Prototype 1:** Frame-by-frame analysis of video segments.
 - o **Prototype 2:** Use of multiple models to detect different types of deepfake manipulations.



Technologies Used:

- **Programming Languages:** Python
- **Frameworks:** TensorFlow, PyTorch
- **Models:** CNNs, RNNs (ongoing exploration)
- **Libraries:** OpenCV, NumPy, SciPy,

System Requirement

Software Requirements

Operating System	:	Windows 11/10 (32-bit or 64-bit)/ Linux
Development Tools	:	Python, TensorFlow, PyTorch
Addition Libraries	:	OpenCV, NumPy, StreamLit

Hardware Requirements

Processor	:	Dual Core 2.7 GHz or better
RAM	:	4GB or higher

Timeline

Task	Timeframe	Description
Project Initiation(T0)	February 2025	Research and requirement gathering
Dataset Collection & Preprocessing (T1)	February 2025	Gathering and preprocessing deepfake datasets
Model Development (T2)	March 2025	Implementing CNN-based deepfake detection models
Testing & Optimization(T3)	March 2025	Improving model accuracy and efficiency
UI Development (T4)	April 2025	Creating a user-friendly detection interface
Final Testing & Report Submission (T5)	End Semester	Comprehensive evaluation and documentation

References

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